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Parameter Efficient Adaptation of Vision Large Language Models for Fisheye Object Detection

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ABSTRACT Fisheye cameras are used for coverage in transportation systems and vehicles, yet distortions pose challenges to object detection algorithms. In this paper, Radial-Aware LoRA (RA-LoRA) is proposed, a distortion-aware parameter-efficient adaptation method that modulates low-rank updates based on radial position to address the spatially varying nature of fisheye distortion. RA-LoRA fine-tuning of Florence-2-large with only 0.5% additional parameters achieves a 181% relative improvement in mAP@50 over zero-shot inference, outperforming fully trained CNN detectors in fisheye imagery. Detection performance is assessed in all five object classes of the FishEye8K dataset (bike, bus, car, pedestrian, and truck). The results show that Florence-2-large with RA-LoRA achieves a mean average precision (mAP) of 0.724, outperforming YOLO26x (mAP: 0.581) and YOLO26 l (mAP: 0.605). Although the model requires approximately 4.5× more memory than YOLO26, the RA-LoRA adapters represent only 0.5% of the total parameters and can be merged into the base model at inference time with no additional overhead. This makes the proposed approach particularly suitable for server-side processing, offline traffic analysis, post-incident auditing, and high-accuracy surveillance applications where detection reliability outweighs the need for real-time embedded deployment.

INDEX TERMS Fisheye cameras, object detection, parameter-efficient fine-tuning, RA-LoRA, Florence-2, vision large language models.

I. INTRODUCTION

Fisheye cameras provide ultra-wide-angle coverage of complex environments such as urban intersections, public spaces, and vehicle surroundings, making them increasingly attractive for autonomous transportation, surround-view perception, and traffic monitoring [1]. A single fisheye sensor can replace several rectilinear cameras while reducing system cost, wiring complexity, and form factor, which is particularly valuable for embedded automotive platforms and smart-city infrastructure [2]. In autonomous driving, fisheye coverage mitigates hidden spots and supports near-field obstacle detection, whereas in traffic analytics it enables wide-area monitoring of vehicles and pedestrians from a single vantage point. However, the optical characteristics that yield this wide field of view also introduce strong geometric distortion that challenges computer vision pipelines designed for standard rectilinear imagery [3].

Fisheye distortion is spatially non-uniform: objects near the optical center remain approximately rectilinear, whereas peripheral regions exhibit increasing radial compression and shape deformation [4], [5], [6]. This warping alters the apparent scale, aspect ratio, and contour of objects, leading to systematic localization errors and class confusion. Detectors trained on rectilinear data therefore exhibit a measurable performance drop when applied to fisheye imagery, since their convolutional priors and anchor distributions assume a perspective projection that no longer holds [7], [8].

The degradation is most pronounced for small and distant objects, where pixel support is already limited and peripheral compression further reduces discriminative cues.

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Reliable detection in these regions is critical for safety-relevant applications such as collision avoidance, where missed pedestrians or cyclists at the image periphery can have serious consequences [9]. Existing approaches address this challenge mainly through geometric data augmentation or full retraining on fisheye corpora, yet a systematic comparison of recent detection paradigms on standardized fisheye benchmarks remains scarce [10].

Vision-language large models (VLLMs) such as Florence-2-large are pre-trained on large-scale multimodal corpora and exhibit strong zero-shot transfer across diverse vision tasks, reducing the dependence on task-specific labeled data [11], [12]. Their attention-based representations encode geometric and semantic priors that, in principle, can compensate for fisheye distortion more effectively than purely convolutional detectors [13], [14]. To date, however, the behavior of Florence-2-large on geometrically distorted imagery has received limited attention, and its trade-off against fast convolutional detectors in fisheye settings has not been quantified.

A controlled evaluation of YOLO variants against Florence-2-large on a common fisheye benchmark is therefore necessary to characterize their real-world applicability. YOLO detectors are optimized for inference speed and direct anchor-based prediction, whereas VLLMs rely on cross-modal reasoning over learned visual tokens; the resulting trade-offs in accuracy, distortion robustness, and computational footprint directly determine which family is suitable for resource-constrained automotive deployment.

This paper presents a controlled comparison of Florence-2-large and YOLO26 on fisheye traffic imagery, jointly evaluating detection accuracy across distortion levels and environmental conditions, computational efficiency in terms of inference latency and memory footprint, and per-class performance over all five object classes of the FishEye8K dataset (bike, bus, car, pedestrian, and truck), thereby exposing the accuracy cost trade-offs relevant for real-time and embedded deployment.

Although YOLO26 and Florence-2-large differ in architecture and training paradigm, their comparison is both relevant and necessary. YOLO26 represents the current standard for real-time CNN-based detection on embedded platforms, while Florence-2-large exemplifies the emerging class of VLLMs that uses large-scale pre-training. This direct evaluation on the same FishEye8K benchmark quantifies the practical trade-offs in accuracy, distortion robustness, and computational efficiency between these two dominant modern detection paradigms.

The main contributions of this paper are:

- A distortion-aware adaptation method, termed Radial-Aware LoRA (RA-LoRA), is proposed. It modulates LoRA updates based on radial position encoding, enabling the model to learn position-dependent feature transformations that explicitly address the spatially varying nature of fisheye distortion.
- RA-LoRA is integrated into a VLLM (Florence-2-large) to adapt pre-trained geometric priors for fisheye imagery. The method requires only 0.5% additional parameters while maintaining full compatibility with the original transformer architecture and task-prompting pipeline.
- A systematic comparison of six detection architectures is conducted on the FishEye8K dataset. This study highlights the inherent robustness of self-attention mechanisms to spatially non-uniform distortion compared to conventional convolutional detectors.

By quantifying the performance efficiency trade-offs of modern detectors on fisheye imagery and introducing a distortion-aware adaptation strategy, this work contributes toward more reliable computer vision systems for safety-critical applications under non-rectilinear projection.

II. METHODOLOGY

This study conducts a systematic comparison between convolutional neural network (CNN) detectors and VLLMs for object detection in fisheye imagery. All models were trained and evaluated on the same FishEye8K dataset split (70% training, 30% testing), with images resized to 1280×1280 pixels and no additional data augmentation applied during evaluation.

A. MODEL SELECTION AND JUSTIFICATION

YOLO26 family (n, l, x variants) represents the current state-of-the-art in real-time CNN-based detection, while Florence-2-large serves as a representative VLLM. Although these models differ in architecture and training paradigm, their comparison is essential because both are widely used in modern perception systems and offer distinct trade-offs in accuracy, robustness to distortion, and computational cost. YOLO26 is optimized for speed on embedded hardware, whereas Florence-2-large leverages large-scale pre-training and cross-modal knowledge to potentially better handle geometric distortions. This direct comparison quantifies the practical benefits and limitations of each approach on the same fisheye benchmark. Training followed standard protocols for each model family. YOLO26 models were trained for 100 epochs with MuSGD, per Ultralytics recommendations for detection. DETR and RT-DETR were trained for 50 epochs with AdamW, following their standard configurations. Florence-2-large was fine-tuned for 20 epochs with AdamW and a low learning rate, as is typical for parameter-efficient adaptation of large vision-language models to avoid catastrophic forgetting. These settings ensure performance differences mainly reflect architectural strengths rather than arbitrary hyperparameters. Fig. 1 illustrates the overall pipeline.

B. IMPLEMENTATION DETAILS

For Florence-2-large, object detection is performed using the unified `[DETECTION]` prompt. The model generates a sequence of bounding boxes and class labels in COCO format. During RA-LoRA fine-tuning, only the vision encoder layers

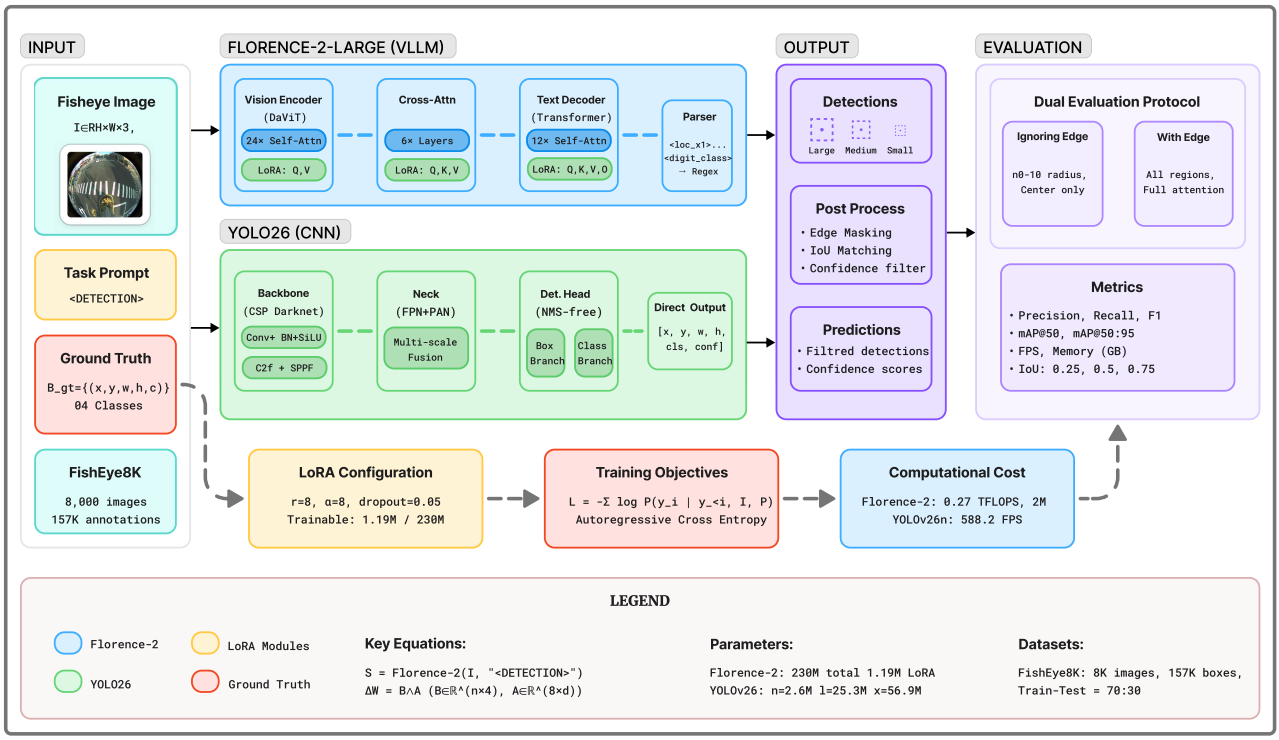


FIGURE 1. Object Detection Pipeline from Ground-truth to Final Output with Florence-2-large and YOLO26 Training Methodology.

receive low-rank adapters. Input images are processed at their native resolution without letterboxing. Inference uses greedy decoding with a maximum of 100 tokens.

All YOLO26 models were trained using the official Ultralytics framework with default hyperparameters (image size 1280, batch size 16, 100 epochs, SGD optimizer with initial learning rate 1×10^{-3}).

DETR was implemented with a ResNet-50 backbone, and RT-DETR with its official ResNet-50 implementation; both were trained for 50 epochs using their standard configurations (batch size 8, AdamW). RT-DETR was evaluated at both 640×640 and 1280×1280 input resolutions.

Training was performed on 4 NVIDIA A100 GPUs. All models were evaluated using the official COCO evaluation protocol at IoU thresholds of 0.25, 0.5, and 0.75. Edge-aware evaluation was conducted by excluding objects whose bounding box centers lie within 5% of the image border.

C. LORA FOUNDATION FOR DOMAIN ADAPTATION

Full fine-tuning of a VLLM with hundreds of millions of parameters is computationally prohibitive. Low-Rank Adaptation (LoRA) addresses this by injecting trainable rank decomposition matrices into the transformer layers, enabling fine-tuning with a reduced parameter count. Our proposed RA-LoRA builds upon this foundation by introducing radial-aware modulation.

Florence-2-large operates using a task-formulation prompt. For object detection, the '[DETECTION]' prompt is used. The model outputs a unified sequence containing bounding-box coordinates and class labels. For an image I, the model generates a sequence S:

$$\hat{y} = f_\theta\big([\texttt{DETECTION}],\, x\big) \tag{1}$$

where $x \in \mathbb{R}^{H\times W\times 3}$ denotes the input fisheye image, [DETECTION] is the Florence-2 task prompt token that conditions the model on the object detection task, f_θ is the Florence-2-large backbone parameterized by θ, and $\hat{y}$ is the predicted set of bounding boxes and class labels.

D. RADIAL-AWARE LORA (RA-LORA)

Standard LoRA applies uniform low-rank updates across all spatial positions, which ignores the radially increasing distortion characteristic of fisheye images. In equidistant fisheye projection models, the radial distance in the image plane is approximately linear with the incident angle θ ($r = f\theta$), resulting in minimal distortion near the optical center and progressively stronger compression and shape deformation toward the periphery. To explicitly account for this geometry, RA-LoRA modulates the low-rank update strength according to the normalized radial distance from the image center.

For each spatial token i, the effective weight is computed as

$$W_{\text{eff}}^{(i)} = W_0 + \alpha(\hat{r}_i) \cdot BA, \tag{2}$$

where the radial modulation factor is

$$\alpha(\hat{r}_i) = 1 + \lambda \cdot \hat{r}_i, \quad \hat{r}_i = \frac{\sqrt{(x_i - c_x)^2 + (y_i - c_y)^2}}{r_{\max}}. \tag{3}$$

This linear scaling directly mirrors the first-order approximation of radial distortion in equidistant fisheye lenses. The modulation is applied consistently to the low-rank update BA

TABLE 1. Model Architecture and Training Configuration

Feature	Florence-2-large	YOLO26	DETR	RT-DETR
Architecture				
Type	Transformer	CNN	Transformer	Transformer
Backbone	DaViT	NMS-free CNN	ResNet-50	ResNet-50
Parameters	230M	2.4M–55.6M	41M	42M
Input	Image+Text	Image	Image	Image
Training Configuration				
LoRA rank (r) / α	8 / 8	N/A	N/A	N/A
LoRA dropout	0.05	N/A	N/A	N/A
Learning rate	4e-6	1e-3	1e-4	1e-4
Batch size	16	16	8	8
Epochs	20	100	50	50
Optimizer	AdamW	MuSGD	AdamW	AdamW

in *every* transformer layer of the vision encoder. Such uniform application across all layers ensures that distortion-aware adaptation propagates through the entire feature hierarchy. The hyperparameter $\lambda = 1.0$ and rank $r = 8$ were used in all experiments. The radial coordinate $\hat{r}_i$ is computed once per image from patch coordinates and passed through the forward pass, requiring no additional learnable parameters.

Geometry-aware methods like DepthFisheye and distortion-aware adapters have been proposed for fisheye imagery. RA-LoRA instead directly integrates radial position encoding into the LoRA update of a VLLM. For each spatial token i, the low-rank update is scaled by

$$\alpha(\hat{r}_i) = 1 + \lambda \cdot \hat{r}_i, \tag{4}$$

where $\hat{r}_i$ is the normalized radial distance from the image center. This yields stronger adaptation in peripheral regions, where fisheye distortion is greatest, while preserving the original weights near the optical center. Unlike prior methods, RA-LoRA needs neither depth estimation nor transformer architectural changes, maintaining full parameter efficiency and compatibility with the Florence-2 pipeline.

III. RESULTS

For Florence-2-large, the experiments are conducted in two modes: 1) Zero-shot, using the base pre-trained model, and 2) RA-LoRA, where the model is adapted using the training split. All YOLO26 models are evaluated in both zero-shot mode (using COCO pre-trained weights) and after training on the FishEye8K training split following their standard training protocols. Table 1 presents a comparison of the architectural differences between Florence-2-large and YOLO26.

A. USED DATASET

FishEye8K [15] is a benchmark dataset for fisheye camera object detection containing 8,000 images with 157,000 bounding-box annotations in five classes: bike, bus, car, pedestrian, and truck. All five classes are used in this work for both training and evaluation, with no class excluded from the experimental protocol. The dataset is collected from 18 fisheye traffic surveillance cameras in Taiwan, at resolutions of 1080 × 1080 and 1280 × 1280 capturing various traffic conditions, illumination levels, and camera angles. The images are split into training and test sets with a 70:30 ratio, ensuring that frames from the camera appear in one set to avoid bias.

TABLE 2. Primary Metrics Comparison for Florence-2-Large

Metric	Ignoring Edge			Considering Edge		
	@0.25	@0.5	@0.75	@0.25	@0.5	@0.75
Precision	0.9853	0.8659	0.4607	0.5126	0.4488	0.2387
Recall	0.8240	0.7242	0.3853	0.8274	0.7243	0.3853
F1 Score	0.8974	0.7887	0.4196	0.6330	0.5542	0.2948

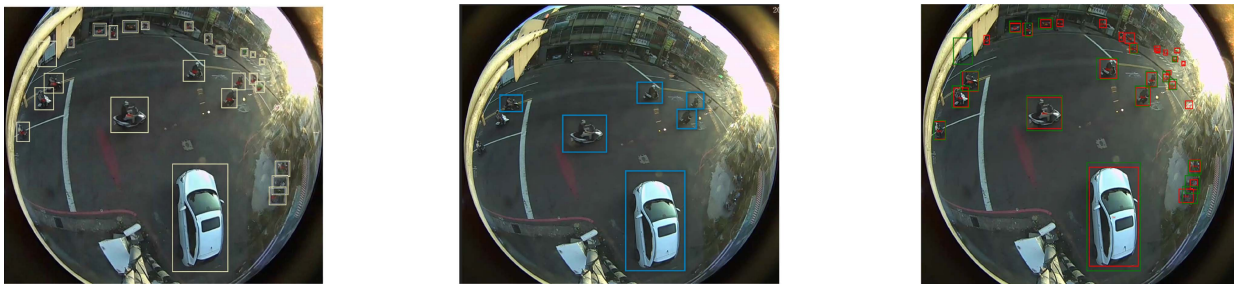
B. EVALUATION METRICS

This comparison examines metrics including detection, precision, recall, and F1 score to assess each model. We used metrics calculated at intersection over union (IoU) thresholds of 0.25, 0.5, and 0.75. The F1 score is computed as the harmonic mean of precision and recall. False discovery rate (FDR) and the false negative rate (FNR) are used to quantify false positives and false negatives. Average Precision (AP) is computed at multiple IoU thresholds, including AP@0.25, AP@0.5, and AP@0.75. All metrics are evaluated under two conditions: ignoring edge regions and considering edge regions of the fisheye image. In addition, an institutional GPU cluster with 4 NVIDIA A100 GPUs is used for training.

C. EXPERIMENTAL RESULTS

Fig. 2 presents a sample comparison of detection results between YOLO26x and Florence-2-large. This demonstrates a superior ability of Florence-2-large to detect edges, despite the lower GPU efficiency. As presented in Table 2, the primary evaluation metrics demonstrate the Florence-2-large VLLM precision drop when edge information is considered. Florence-2-large detects valid but unlabeled objects near image borders, which contributes to lower precision scores despite visually correct detections. When edges are ignored, the model achieves high precision (0.9853 at IoU@0.25) and recall (0.8240), leading to a strong F1 score of 0.8974. However, accounting for edges results in lower precision (0.5126 at IoU@0.25), reflecting the increased challenge of detecting objects with distorted boundaries typical of fisheye imagery. Although recall remains stable in both conditions, the consistent decline in the F1 score highlights the degradation in localization precision near the edges of the object. The substantial drop in precision when edge regions are considered is primarily attributable to missing ground-truth annotations rather than spurious detections by the model. Qualitative analysis shows that Florence-2-large correctly identifies many objects near the image periphery that were not annotated in the FishEye8K test set. Consequently, the reported precision underestimates the true performance of the model in distorted edge regions, while recall remains stable as the model successfully detects most annotated objects.

As shown in Table 3, the error metrics further emphasize the degradation of the detection consistency when edges are



(a) Original Image (b) Object Detection using YOLO26x (c) Object Detection using Florence-2-large

FIGURE 2. YOLO26x and Florence-2-large Comparison for Detecting Objects in Fisheye Images: Green Boxes as Ground Truth.

TABLE 3. Error Metrics Comparison (FDR and FNR)

Metric	Ignoring Edge			Considering Edge		
	@0.25	@0.5	@0.75	@0.25	@0.5	@0.75
FDR	0.0147	0.1341	0.5393	0.4874	0.5512	0.7613
FNR	0.1760	0.2758	0.6147	0.1726	0.2757	0.6147

TABLE 4. Average Precision Metrics Comparison

Metric	Ignoring Edge	Considering Edge
Mean IoU (GT objects)	0.5997	0.5910
AP@0.25	0.87	0.55
AP@0.5 (AP50)	0.76	0.50
AP@0.75 (AP75)	0.41	0.26
mAP@50 (AP50)	0.76	0.50
mAP@50:95	0.40	0.32

TABLE 5. RT-DETR Performance on FishEye8K At Two Input Resolutions

Input size	Prec.	Recall	F1	mAP@50	mAP@50-95
640 × 640	0.567	0.415	0.479	0.457	0.262
1280 × 1280	0.706	0.493	0.581	0.571	0.335

TABLE 6. Performance Metrics Comparison for All Models

Metric	DETR	RT-DETR	YOLO26n	YOLO26l	YOLO26x	Florence
Precision	0.362	0.706	0.615	0.685	0.666	**0.866**
Recall	0.466	0.493	0.351	0.532	0.519	**0.724**
F1 Score	0.408	0.581	0.449	0.599	0.583	**0.789**
mAP@50	0.253	0.571	0.390	0.605	0.581	**0.724**
mAP@50-95	0.107	0.335	0.220	0.359	0.357	**0.401**
FPS	52.9	12.0	588.2	135.1	89.3	0.27

considered. The FDR exhibits an increase from 0.0147 to 0.4874 at IoU@0.25, indicating that nearly half of the detected objects near the image borders are false positives. Similarly, FDR continues to increase in higher IoU thresholds (up to 0.7613 at IoU@0.75), suggesting reduced boundary precision under fisheye distortions. In contrast, the FNR remains stable in both configurations, ranging from 0.17 to 0.61, implying that the model maintains a consistent level of sensitivity even in areas distorted by edges. Object recall is not affected by edge inclusion, but precision deteriorates.

Table 4 presents the comparison of the average precision metrics in both evaluation settings. The mean IoU shows a marginal decline from 0.5997 to 0.5910 when edge regions are included, indicating a slight reduction in overall localization accuracy. Despite this, the Average Precision (AP) values at multiple IoU thresholds remain consistent in both configurations, when ignoring edge regions, AP@0.25 and AP@0.5 achieve 0.87 and 0.76, respectively. Including edges reduces these to 0.55 and 0.50. The higher IoU threshold (AP@0.75) exhibits a drop to 0.26, reflecting the increased difficulty in achieving precise boundary alignment under fisheye distortion. The COCO-style mAP@50:95 decreases from 0.40 to 0.32 when edges are considered, indicating that localization is more sensitive to peripheral distortions.

D. OVERALL MODEL COMPARISON

For comparison, we evaluate two transformer-based detectors: the original DETR [16] and RT-DETR [17], a strong modern real-time transformer detector. As shown in Table 6, Florence-2-large achieves the best precision (0.866), recall (0.724), and F1 score (0.789), outperforming all baselines. RT-DETR substantially improves over the original DETR, raising mAP@50 from 0.253 to 0.571 and precision from 0.362 to 0.706 while running at 12 FPS, and is competitive with the larger YOLO26 variants (YOLO26x mAP@50 0.581, YOLO26 l mAP@50 0.605). Nevertheless, all convolutional and transformer baselines remain below Florence-2-large with RA-LoRA on fisheye imagery, highlighting the challenge of geometric distortion for non-specialized models. Florence-2-large records the highest mAP@50 (0.724) and mAP@50-95 (0.401), surpassing YOLO26x by ∼ 24% and YOLO26l by ∼20% in mAP@50. However, its inference speed is only 0.27 FPS, compared to 588.2 FPS (YOLO26n), 135.1 FPS (YOLO26 l), and 89.3 FPS (YOLO26x).

Following the original FishEye8K benchmark, RT-DETR was additionally evaluated at the lower 640 × 640 input resolution. As reported in Table 5, the higher 1280 × 1280 resolution improves all metrics (e.g., mAP@50 from 0.457 to 0.571), since finer spatial detail aids the detection of small and peripherally distorted objects.

E. ABLATION STUDY: ZERO-SHOT VS. DOMAIN-TUNED PERFORMANCE

To quantify the benefit of domain adaptation, all models are evaluated in both zero-shot and trained/fine-tuned configurations. Table 7 presents this ablation study comparing pre-trained models (using COCO weights for YOLO and the base Florence-2-large checkpoint) against models adapted to

TABLE 7. Zero-Shot Vs. Trained/Fine-Tuned Performance

Model	Mode	Prec.	Recall	F1	mAP@50	mAP@50-95
YOLO26n	Zero-shot	0.624	0.124	0.207	0.124	0.108
	Trained	0.615	0.351	0.449	0.390	0.220
YOLO26l	Zero-shot	0.557	0.309	0.397	0.309	0.270
	Trained	0.685	0.532	0.599	0.605	0.359
YOLO26x	Zero-shot	0.588	0.312	0.408	0.312	0.277
	Trained	0.666	0.519	0.583	0.581	0.357
Florence-2-large	Zero-shot	0.595	0.257	0.359	0.257	0.229
	Vanilla LoRA	0.84	0.702	0.765	0.703	0.382
	RA-LoRA	**0.866**	**0.724**	**0.789**	**0.724**	**0.401**

TABLE 8. Per-Class Average Precision of Florence-2-Large With RA-LoRA on FishEye8K

Class	AP@50	AP@[.50:.95]
Bus	0.680	0.360
Bike	0.790	0.450
Car	0.850	0.500
Pedestrian	0.610	0.310
Truck	0.690	0.385
Average	**0.724**	**0.401**

the FishEye8K training set. All models show performance degradation in zero-shot mode, confirming that fisheye distortions represent a significant domain shift. Florence-2-large with RA-LoRA achieves the largest gain: mAP@50 increases from 0.257 to 0.724 (a **+181%** relative improvement) while updating only rank-8 matrices with radial modulation. The radial-aware scaling enables stronger adaptation in peripheral regions where distortion is most severe. To isolate the contribution of the radial modulation itself, we further compare against a vanilla LoRA baseline trained with identical rank ($r = 8$), α, learning rate, optimizer, and schedule. Under this matched setting, RA-LoRA improves mAP@50 by approximately 3% (0.703 $\rightarrow$ 0.724) and mAP@50-95 by approximately 5% (0.382 $\rightarrow$ 0.401) relative to vanilla LoRA, confirming that the gain stems from the radial-aware scaling rather than from low-rank adaptation alone. These results validate the distortion aware parameter-efficient tuning for fisheye imagery.

F. PER-CLASS PERFORMANCE AND RADIAL MODULATION STRENGTH

Table 8 reports the per-class average precision of Florence-2-large with RA-LoRA across the five FishEye8K classes. Car (AP@50 0.850) and bike (0.790) are detected most reliably, whereas pedestrian (0.610) is the hardest class, reflecting its small pixel support and frequent appearance in peripheral regions where fisheye compression is most severe. Bus and truck achieve comparable accuracy (0.680 and 0.690), consistent with their visual similarity under distortion.

We further analyze the sensitivity to the radial modulation strength λ, which controls how strongly the low-rank update is amplified toward the image periphery. As reported in Table 9, increasing λ from 0 (equivalent to standard LoRA) to the proposed value $\lambda = 1.0$ raises mAP@50 from 0.703 to 0.724, while mAP@[.50:.95] saturates at 0.401 beyond $\lambda = 0.5$. This

TABLE 9. Sensitivity to the Radial Modulation Strength λ (Florence-2-Large, $r = 8$)

λ	mAP@50	mAP@[.50:.95]
0.0 (standard LoRA)	0.703	0.382
0.5	0.710	0.401
1.0 (proposed)	**0.724**	**0.401**

TABLE 10. Effect of Input Resolution: Proposed RA-LoRA Vs. RT-DETR on FishEye8K

Model	Input size	mAP@50	mAP@[.50:.95]
RA-LoRA (ours)	640 × 640	0.557	0.308
RA-LoRA (ours)	1280 × 1280	**0.724**	**0.401**
RT-DETR	640 × 640	0.457	0.262
RT-DETR	1280 × 1280	0.571	0.335

confirms that radial-aware scaling provides a consistent improvement over uniform LoRA and that $\lambda = 1.0$ offers the best overall accuracy.

Finally, to remain consistent with the original FishEye8K benchmark, we evaluate the proposed RA-LoRA and the RT-DETR baseline at both the 640 × 640 and 1280 × 1280 input resolutions (Table 10). Both models benefit from the higher resolution, with RA-LoRA improving from mAP@50 0.557 to 0.724, as finer spatial detail aids the detection of small and peripherally distorted objects. RA-LoRA outperforms RT-DETR at both resolutions (by 0.100 and 0.153 mAP@50, respectively), confirming that the advantage of radial-aware adaptation holds across input scales rather than at a single setting.

IV. DISCUSSION

Three observations emerge from the empirical results. First, the accuracy gap between transformer-based and CNN-based detectors is most pronounced in the outer image annulus, where radial distortion is largest, and narrows toward the optical center. This pattern supports the hypothesis that global attention better compensates for spatially non-stationary deformations than fixed local receptive fields. Second, the apparent precision drop when including peripheral regions is largely due to annotation incompleteness in the FishEye8K dataset rather than spurious detections. Manual inspection reveals that a substantial fraction of false positives correspond to genuine objects missing from the ground truth. Third, the benefit of RA-LoRA saturates beyond rank $r = 8$, indicating that further gains are limited by data quality and long-tailed object-size distribution rather than adapter capacity.

The deployment trade-off remains clear: Florence-2-large with RA-LoRA achieves superior accuracy at the cost of higher computational demand (9.2 GB memory and 0.27 FPS). It is therefore best suited for server-side and offline applications such as smart-city traffic monitoring, forensic video analysis, and high-precision auditing, where detection reliability is prioritized. The trained LoRA adapters (0.5% of total parameters) can be merged into the base model at inference time with no additional overhead, while

YOLO26 remains the practical choice for embedded real-time systems.

V. CONCLUSION

This paper introduces RA-LoRA, a distortion-aware, parameter-efficient adaptation technique for vision-language models in fisheye object detection. By modulating LoRA updates by radial position, RA-LoRA enables Florence-2-large to achieve a 181% relative mAP@50 gain over zero-shot inference while updating only 0.5% of parameters. While the full model is computationally heavy, it is practical for server-side, offline, high-accuracy applications such as smart-city traffic monitoring, forensic analysis, and post-event auditing, where reliability outweighs real-time constraints. RA-LoRA adapters can be merged into the base model at inference with no extra overhead. Future work will distill radial-aware knowledge into lighter architectures for edge deployment. Several directions remain. First, the 9.2 GB memory footprint and 0.27 FPS throughput of Florence-2-large preclude embedded and real-time use, motivating lightweight radial-aware models via knowledge distillation into compact students, plus low-bit quantization (INT8/INT4) and structured pruning of RA-LoRA adapters. Second, RA-LoRA's radial modulation is not limited to equidistant fisheye optics and may extend to other distorted modalities, such as catadioptric, omnidirectional, and panoramic imagery. Third, hardware-aware neural architecture search that jointly optimizes mAP, latency, and memory under radial-aware constraints is a promising path to production-ready fisheye perception for autonomous driving and surround-view monitoring. This study has three main limitations: Florence-2-large's 9.2 GB memory requirement prevents edge deployment; its 0.27 FPS inference speed precludes real-time applications like autonomous vehicles; and missing labels in FishEye8K inflate false discovery rates and underestimate performance in distorted regions. The key theoretical contribution is showing that explicitly encoding geometric distortion into parameter-efficient adaptation substantially improves the robustness of large pre-trained models to non-rectilinear imagery, laying a foundation for geometry-aware adaptation of VLLMs beyond fisheye domains.

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